DO HIGHER PRODUCT RATINGS DRIVE MORE PURCHASES ON AMAZON?

Team D5

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PROBLEM DEFINITION/MOTIVATION

Problem Defination: Our hypothesis is that higher product ratings are associated with more purchases on Amazon.

Business & Marketing Strategy Impact
Understanding the relationship helps sellers decide where to invest, product improvements, promotions, obtaining badges, or generating reviews.

Platform Trust & Consumer Behavior
Ratings influence how customers make purchase decisions. Measuring their true effect validates whether Amazon's rating system is a trustworthy signal of product quality.

Data-Driven Decision Making
Insights from this relationship allow retailers and the platform to optimize recommendation systems, search ranking and pricing strategies based on evidence rather than assumptions.

HYPOTHESES & EXPECTED OUTCOMES

Hypothesis:

- **Null Hypothesis (H_0):** Star ratings have no significant impact on product sales volume/purchases
- **Alternative Hypothesis (H_1):** Higher star ratings are correlated with more purchases (i.e. higher sales)

Need for empirical evaluation:

- Prior studies suggest the rating–sales relationship varies by context
  - product type significantly moderates how ratings influence purchase behavior (ScienceDirect, 2023).
- Confounders like category, review volume, and pricing blur intuition.

Possible Relationship Outcomes:

- **Positive Relationship** - Higher ratings strengthen trust and satisfaction, increasing purchase likelihood.
- **No Relationship** - Sales may be driven more by price, visibility, or promotions than ratings.
- **Negative Relationship** - Highly rated niche or premium products may generate fewer overall sales.

IDEAL EXPERIMENT

Randomization

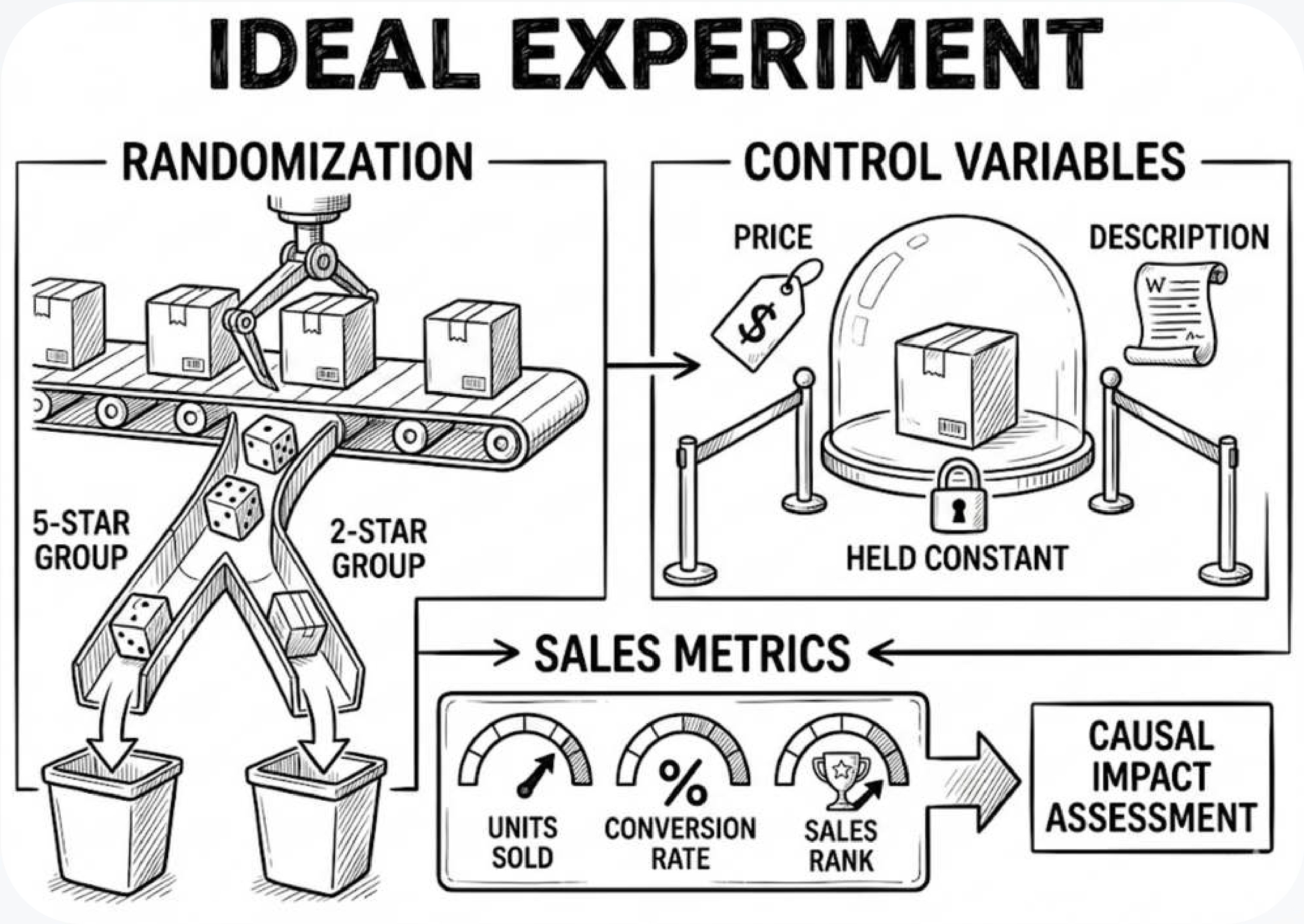
Randomly assign identical products to varying star ratings, ensuring a controlled environment for testing.

Control Variables

Hold all other elements constant, including price and descriptions, to eliminate confounding factors.

Sales Metrics

Measure changes in units sold, conversion rates, and sales ranks to assess the causal impact.



DATA OVERVIEW & SUMMARY STATISTICS

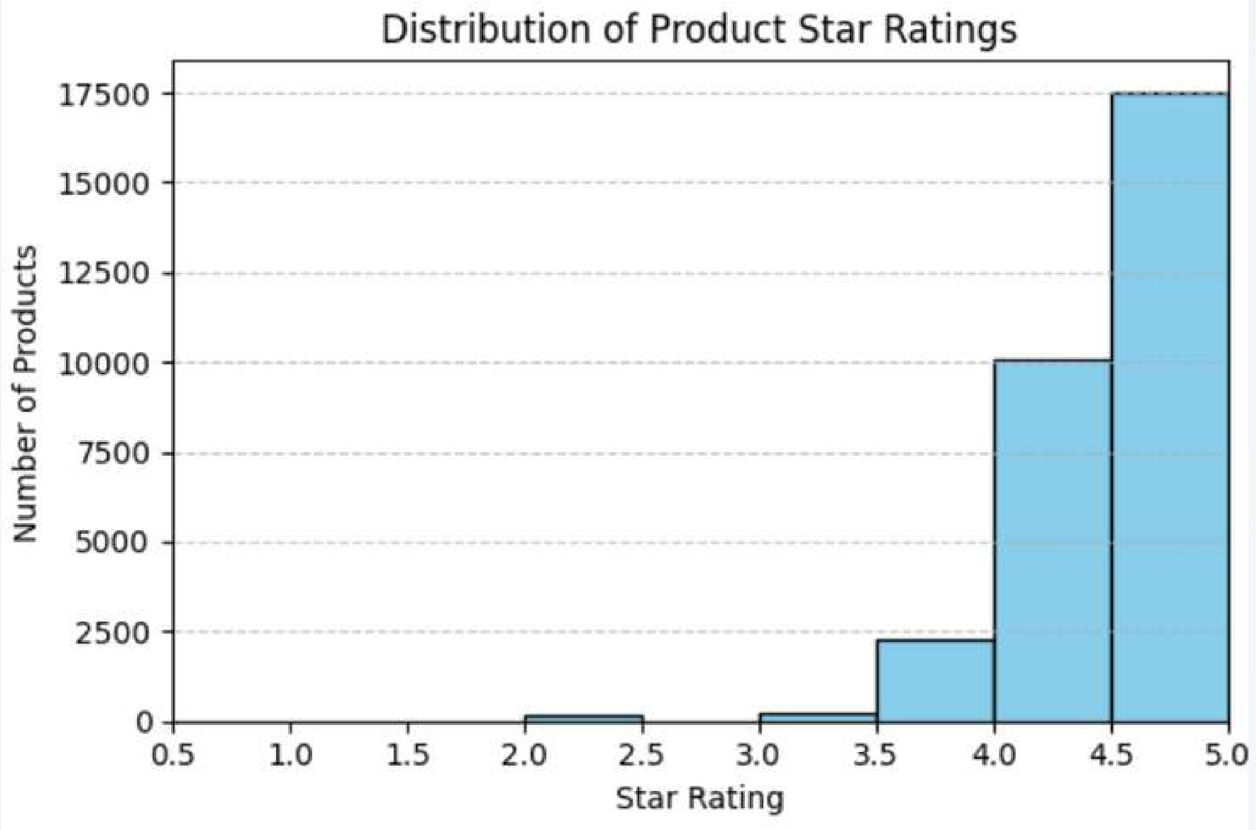
Dataset
Kaggle (Amazon Product Sales Dataset)

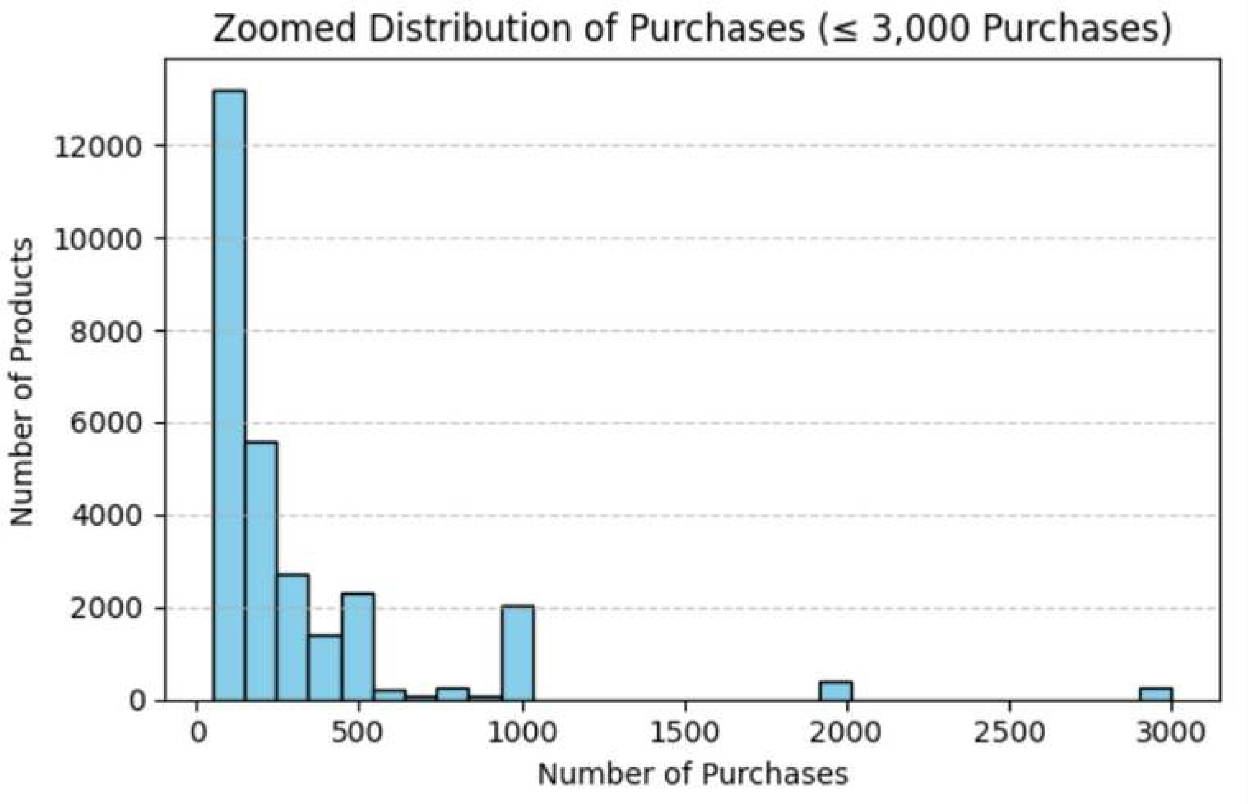
Aggregation
Product-level
(each row = one product)

Variables
Independent: product_rating
Dependent: purchased_last_month

	count	mean	median	std	min	max	nan_count
product_rating	30228	4.439	4.5	0.359	2.0	5.0	0
purchased_last_month	30228	1354.570	200.0	6476.654	50.0	100000.0	0

UNIVARIATE ANALYSIS: RATINGS & PURCHASES





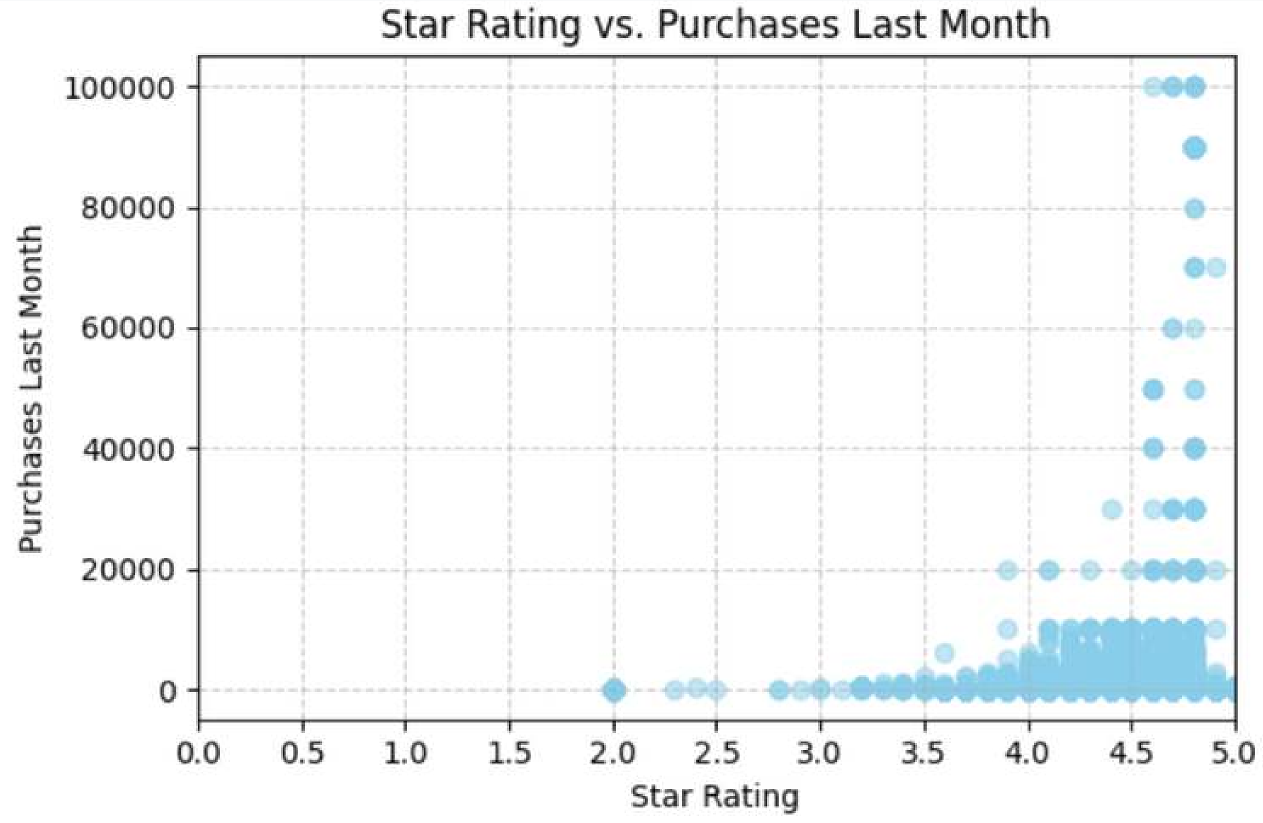
Insights (Ratings):

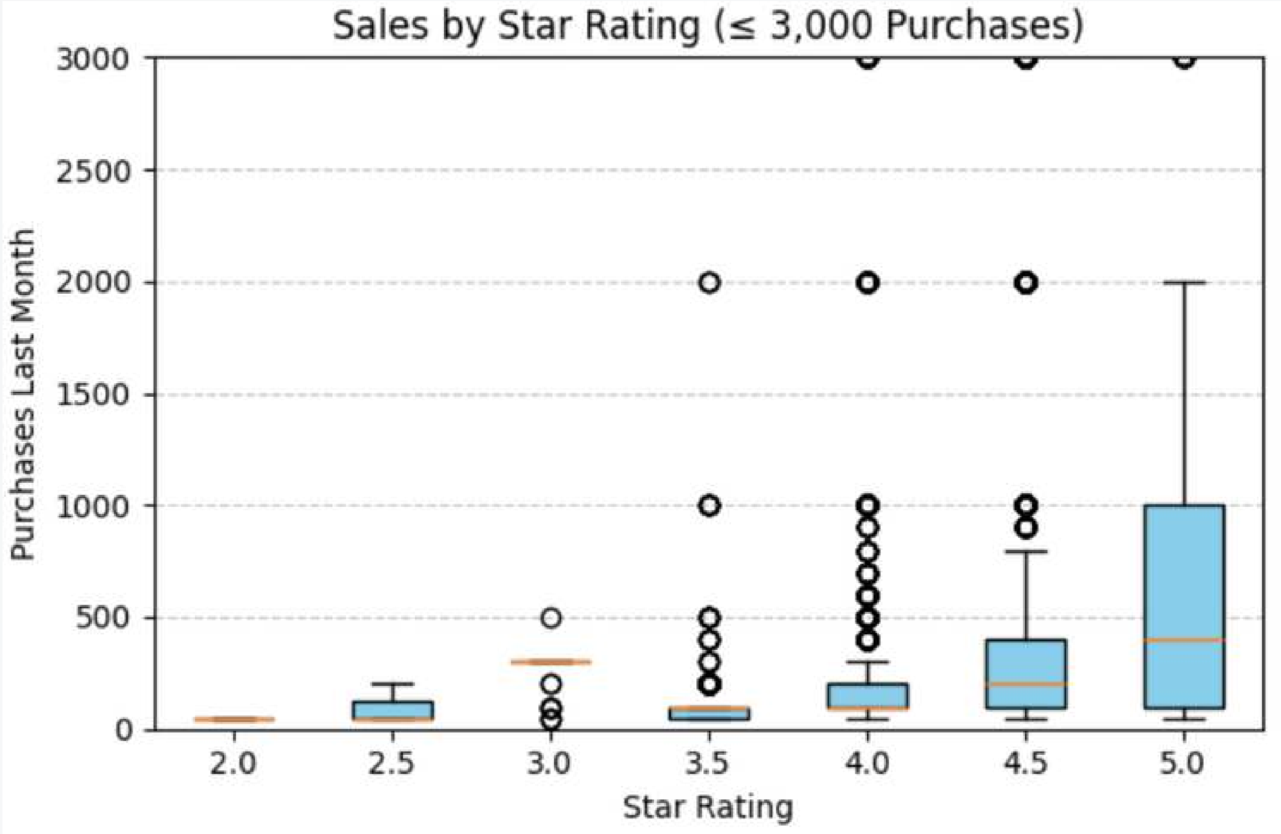
- Left-skewed, concentrated around 4.0–5.0.
- Very few ratings below 2.0 <-> selection bias
- We use 3.0 as the baseline for “low rating”.
  - Supporting evidence:
    - “...average star rating is 4.2 / 5, with well over half of the reviews being five-star ratings” (RetailWire, 2023).
    - “Negative experiences (1-2 stars) are more likely to include written text” (AMALYTIX, 2023).

Insights (Purchases):

- Right-skewed, most products have < 1,000 purchases.

BIVARIATE ANALYSIS: RATINGS VS. PURCHASES





Insights (Scatter Plot):

- higher ratings generally correspond to more purchases.

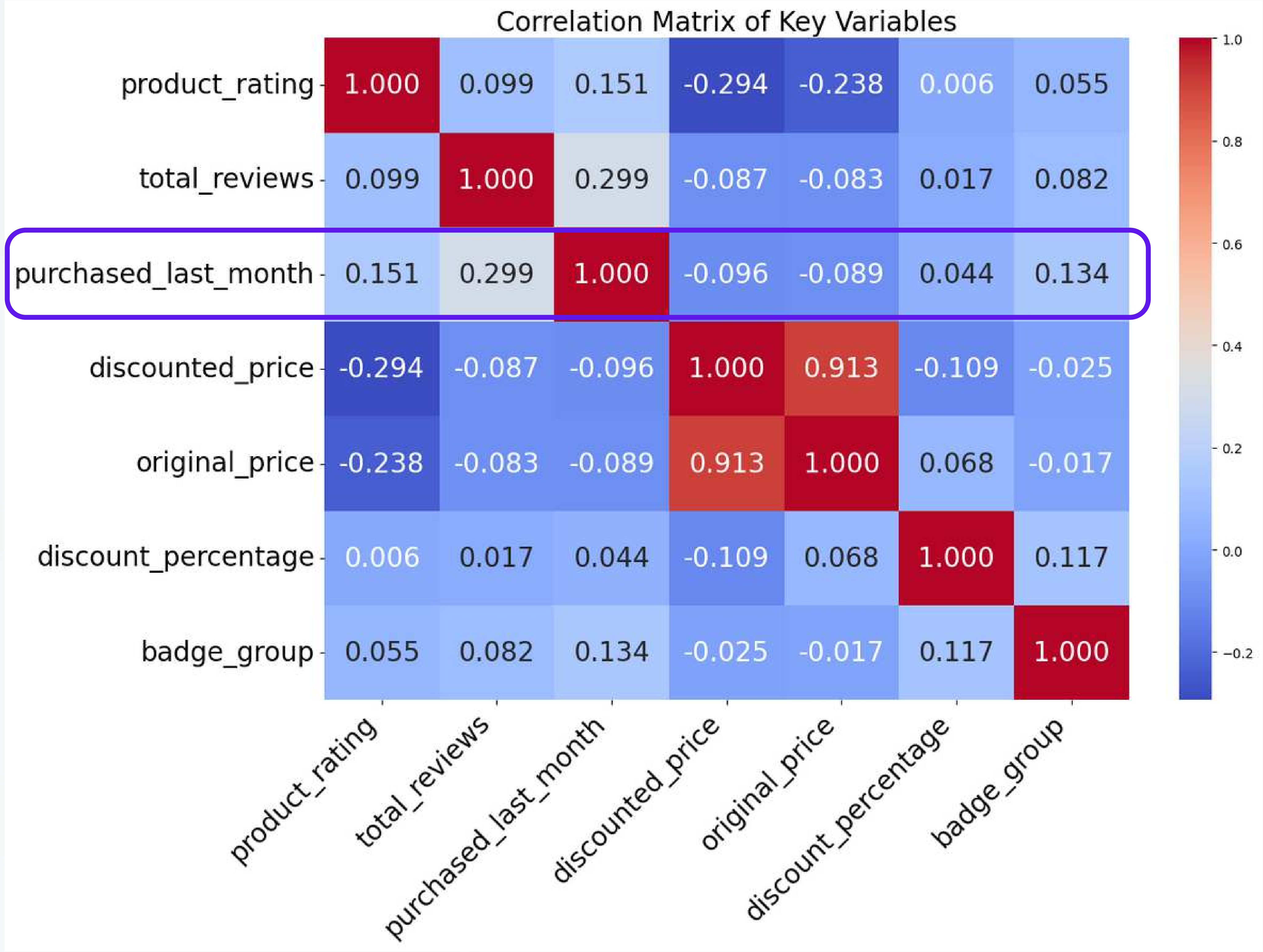
Insights (Box Plot):

- As ratings increase, median purchases generally increase.

Understanding the Variation Behind the Trend:

- High-rating items include both niche and viral products, creating wide variation in purchases/sales.
- A cluster of high-rating, high-sales items hints at product category or badge effects.

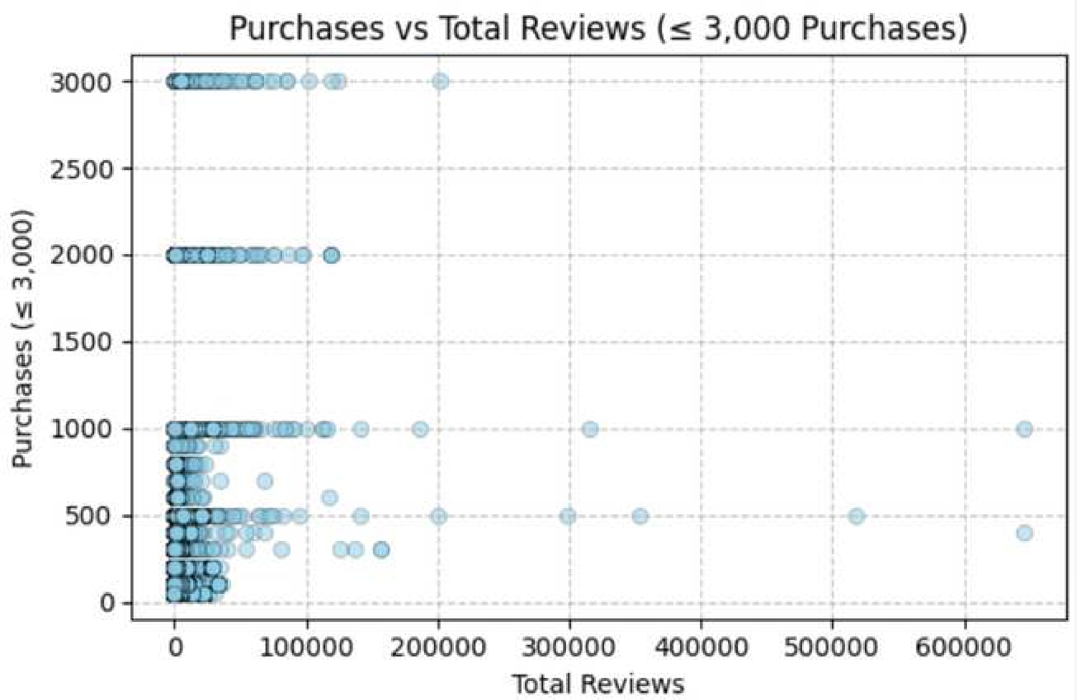
CORRELATION MATRIX & CONFOUNDERS

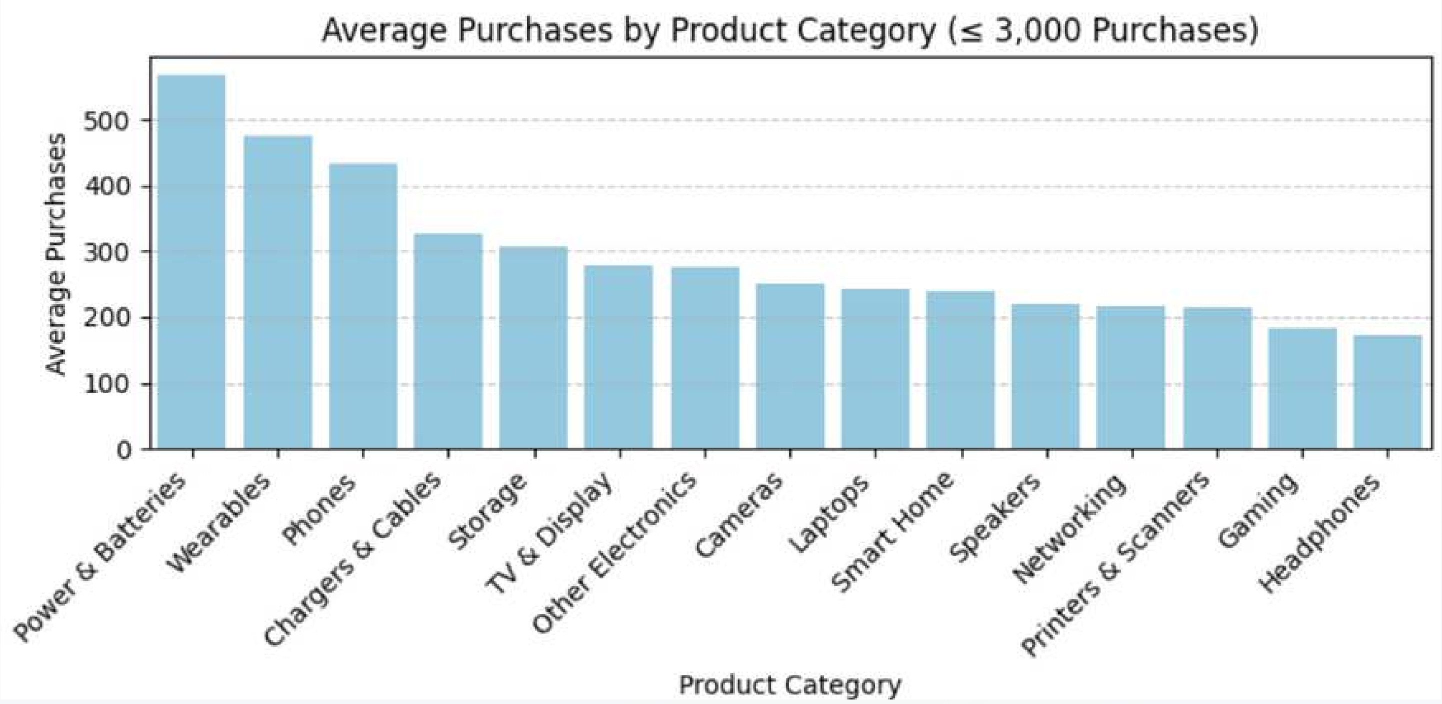


Potential Confounding Variables

- **Total Reviews:** more reviews signal trust/popularity, boosting ratings & purchases
- **Badges (0 = No Badge, 1 = Has Badge):** official badges like Best Seller signal product quality and popularity, directly boosting purchases and potentially influencing ratings
- **Product Category:** different categories attract different levels of consumer demand

CONFOUNDER VISUALIZATIONS

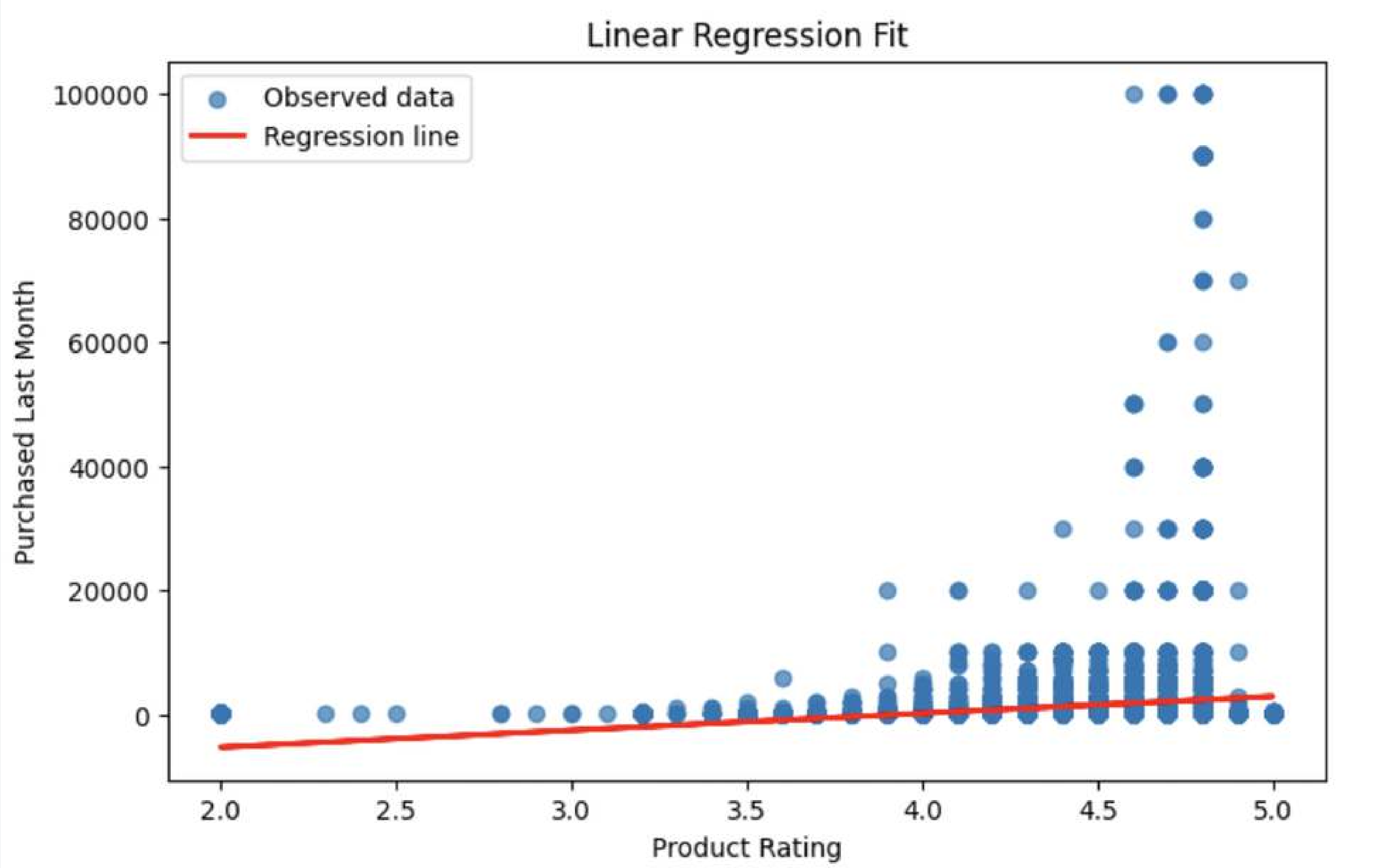




Insights:

1. **Review Variability:**
   - Reviews show only a weak positive link to purchases
   - High review counts do not guarantee high purchases
2. **Badge Influence:**
   - Products with badges have higher median purchases
   - Strong overlap shows badges alone don’t drive purchases
3. **Product Category Effect:**
   - Purchases vary widely by category
   - Power & Batteries, Wearables, and Phones lead demand

SIMPLE REGRESSION

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                            OLS Regression Results
==============================================================================
Dep. Variable:     purchased_last_month   R-squared:                       0.023
Model:                              OLS   Adj. R-squared:                  0.023
Method:                   Least Squares   F-statistic:                     709.0
Date:                  Tue, 02 Dec 2025   Prob (F-statistic):          2.00e-154
Time:                          08:36:02   Log-Likelihood:            -3.0782e+05
No. Observations:                 30228   AIC:                         6.156e+05
Df Residuals:                     30226   BIC:                         6.157e+05
Df Model:                             1
Covariance Type:              nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const          -1.077e+04    456.801    -23.575      0.000    -1.17e+04   -9873.738
product_rating  2731.3088    102.577     26.627      0.000     2530.254    2932.363
==============================================================================
```

This model predicts that an additional unit increase in rating for a product will increase purchases by ~2731.

MULTIVARIABLE REGRESSION

Target: Amount Purchased Last Month

Dependent Variables:

- Product Rating
- Discount Percentage
- Total Reviews
- Product Rating * Sponsored (Interaction Term)
- Badge Group (Baseline: No Badge)
- Sponsored (Baseline: Not Sponsored)
- Has Coupon (Baseline: Has a Coupon)
- Product Category (Baseline: Cameras)

R-Square = 0.307

OLS Regression Results

Dep. Variable:	purchased_last_month	R-squared:	0.307
Model:	OLS	Adj. R-squared:	0.307
Method:	Least Squares	F-statistic:	638.5
Date:	Tue, 02 Dec 2025	Prob (F-statistic):	0.00
Time:	05:46:18	Log-Likelihood:	-3.0262e+05
No. Observations:	30228	AIC:	6.053e+05
Df Residuals:	30206	BIC:	6.055e+05
Df Model:	21		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-1263.5325	488.658	-2.586	0.010	-2221.323	-305.742
product_rating	-79.3979	96.270	-0.825	0.410	-268.091	109.295
discount_percentage	17.5944	2.361	7.453	0.000	12.967	22.222
total_reviews	0.1108	0.002	52.215	0.000	0.107	0.115
int_pr_s	9768.3125	320.296	30.498	0.000	9140.518	1.04e+04
badge_group	1478.8506	141.114	10.480	0.000	1202.261	1755.440
is_sponsored_Sponsored	-3.925e+04	1447.015	-27.125	0.000	-4.21e+04	-3.64e+04
has_coupon_No Coupon	1068.8974	177.449	6.024	0.000	721.090	1416.704
product_category_Chargers & Cables	231.0020	176.825	1.306	0.191	-115.582	577.586
product_category_Gaming	195.9813	277.361	0.707	0.480	-347.658	739.620
product_category_Headphones	318.1925	214.072	1.486	0.137	-101.397	737.782
product_category_Laptops	394.5438	129.677	3.043	0.002	140.371	648.716
product_category_Networking	366.7980	219.873	1.668	0.095	-64.162	797.758
product_category_Other Electronics	667.9808	131.572	5.077	0.000	410.093	925.868
product_category_Phones	-102.6996	136.761	-0.751	0.453	-370.757	165.358
product_category_Power & Batteries	4997.1473	164.870	30.310	0.000	4673.995	5320.300
product_category_Printers & Scanners	482.3145	213.219	2.262	0.024	64.396	900.233
product_category_Smart Home	-2.0858	358.193	-0.006	0.995	-704.159	699.988
product_category_Speakers	372.6366	205.785	1.811	0.070	-30.710	775.983
product_category_Storage	-636.1585	206.361	-3.083	0.002	-1040.634	-231.683
product_category_TV & Display	117.7784	182.356	0.646	0.518	-239.648	475.205
product_category_Wearables	1615.6016	282.058	5.728	0.000	1062.755	2168.448

MULTIVARIABLE REGRESSION

	product_title	product_rating	total_reviews	badge_group	is_sponsored	has_coupon	product_category	discount_percentage
0	BOYA BOYALINK 2 Wireless Lavalier Microphone f...	4.6	375.0	No Badge	Sponsored	Save 15% with coupon	Phones	43.60

Purchases = b0 + b1 (Product Rating) + b2 (Discount Percentage) + b3 (Total Reviews) + b4 (Pr * Sp) + b5 (Badge Group) + b6 (Sponsored) + b7 (Coupon) + b8 (Category)

= -1263.5 - 79.4 (4.6) + 17.5 (43.6) + 0.11 (375) + 9768.3 (4.6 * 1) + 1478.9 (0) - 39250 (1) - 102.7 (1)

Predicted Purchases = 4757

Actual Purchases = 300

95% PI = Prediction ± t * RMSE = 4757 ± 1.96 * 5390 = [-5804, 1532]

TAKEAWAYS

1. Rating does influence purchases, although in different amounts depending on sposored subgroup.

2. The model has a low prediction power, this may be due to issues in the data like heteroscedasticity or non-linear relationships between the target variable and dependent variables.

3. Other key features could be more influential to purchases. Feature engineering may be needed for the better predictive model.

THANK YOU

References:

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- RetailWire. (2023, August 15). Do five-star ratings systems have a “positivity” problem? https://retailwire.com/discussion/do-five-star-ratings-systems-have-a-positivity-problem/
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